

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12 Mathematics Extension 2

HSC Course

Assessment 2

March, 2015

Time allowed: 70 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A set of Standard Integrals is provided at the rear of this Question Booklet, and may be removed at any time.

Section 1 Multiple Choice
Questions 1-5
5 Marks

Section II Questions 6-9
40 Marks

Section I

5 marks

Attempt Questions 1-5

Use the multiple choice answer sheet for Questions 1 – 5.

1. A square root of $8 + 6i$ is :

- (A) $3 - i$ (B) $5 - 3i$
(C) $-3 - i$ (D) $-3 + i$

2. The equation of a curve is given by $x^2 + xy + y^2 = 9$. Which of the following expressions will provide the value of $\frac{dy}{dx}$ at any point on the curve?

- (A) $\frac{-2x - y}{2y}$ (B) $\frac{-2x - y}{x + 2y}$
(C) $\frac{-2x + y}{2y}$ (D) $\frac{-2x + y}{x + 2y}$

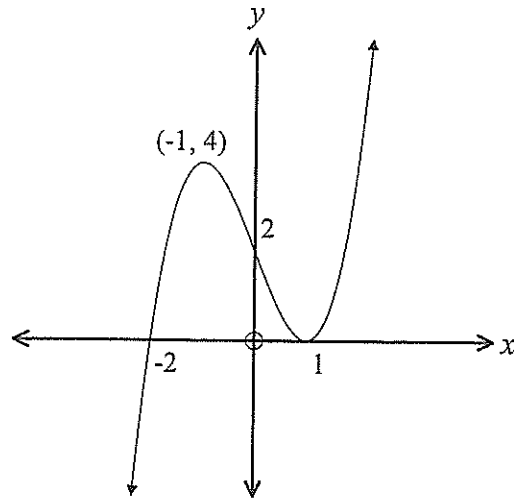
3. The equation of an hyperbola is given by $9x^2 - 4y^2 = 36$. The foci and the directrices of this hyperbola are:

- (A) $(\pm\sqrt{13}, 0)$ and $x = \pm\frac{4\sqrt{13}}{13}$.
(B) $(0, \pm\sqrt{13})$ and $x = \pm\frac{4\sqrt{13}}{13}$.
(C) $(\pm\sqrt{13}, 0)$ and $y = \pm\frac{4\sqrt{13}}{13}$.
(D) $(0, \pm\sqrt{13})$ and $y = \pm\frac{4\sqrt{13}}{13}$.

4. The area bounded by the curves $y = x^2$ and $x = y^2$ is rotated about the x - axis. The volume of the solid of revolution formed in cubic units is:

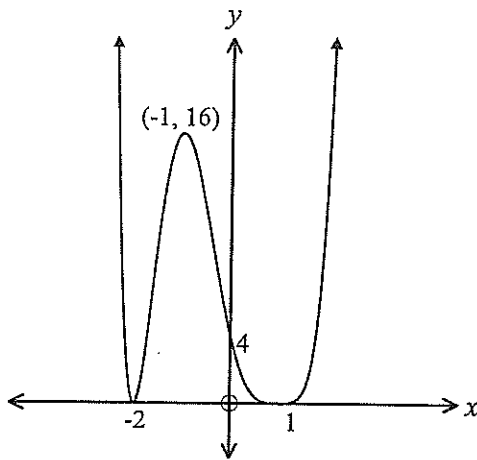
- (A) $\frac{9\pi}{70}$ (B) $\frac{3\pi}{10}$
(C) $\frac{7\pi}{10}$ (D) $\frac{3\pi}{2}$

5. The graph of the function $y = f(x)$ is drawn below:

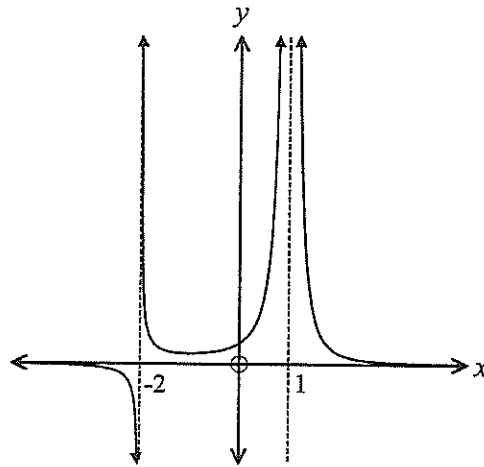


Which of the following graphs best represents the graph $y = \sqrt{f(x)}$?

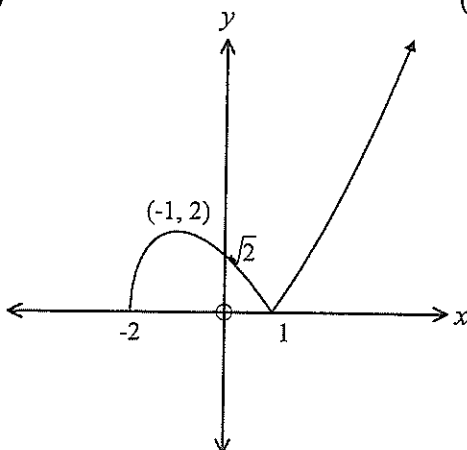
(A)



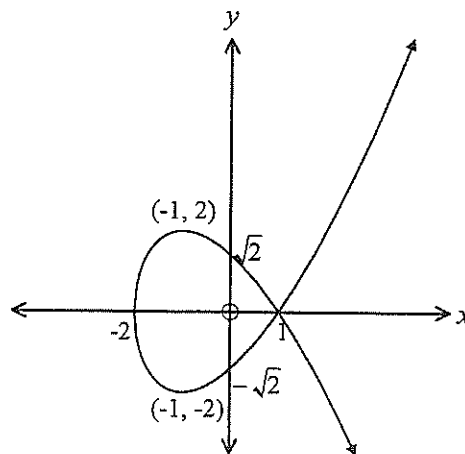
(B)



(C)



(D)



End of Section I

Section II

Total marks (40)

Attempt Questions 6 - 9

Question 6 (10 marks)

Marks

a) An ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{2} = 1$

(i) Show that the equation of E can be written in the parametric form

2

$$x = 2\cos\theta, y = \sqrt{2}\sin\theta$$

(ii) Assuming the perimeter of E is given by the formula

2

$$p = 2 \int_0^\pi \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta,$$

$$\text{show that } p = 2\sqrt{2} \int_0^\pi \sqrt{2 - \cos^2\theta} d\theta$$

b) (i) If $w = \frac{1+i\sqrt{3}}{2}$ show that $w^3 = -1$

1

(ii) Hence calculate w^{12}

1

(iii) Find all the cube roots of -1 , both Real and Complex.

2

c) Given that one root of the equation $x^4 - 5x^3 + 5x^2 + 25x - 26 = 0$ is $3 + 2i$, solve the equation.

2

Question 7 (10 marks) Start a new page

- a) If $f(x) = -x^2 + 7x - 10$, on separate diagrams and without using calculus, sketch the following graphs, indicating the intercepts with the axes and any asymptotes for each sketch:
- (i) $y = f(x)$ 1
 - (ii) $y = |f(x)|$ 2
 - (iii) $y = \frac{1}{f(x)}$ 2
 - (iv) $y = -f(x + 2)$ 2
- b) Find all the roots of $18x^3 + 3x^2 - 28x + 12 = 0$, given that two roots are equal. 3

Question 8 (10 marks) Start a new page

- a) $z_1 = 1 + i\sqrt{3}$ and $z_2 = 1 - i$ are two complex numbers 2
- find $\frac{z_1}{z_2}$ in modulus-argument form
- b) Given that the Argand Diagram for $|z - 2| + |z - 4| = 10$ is an ellipse,
- (i) Find the co-ordinates of the centre of this ellipse and the lengths of the major and minor axes 3
 - (ii) On an Argand Diagram, show the region for which z satisfies the inequalities 3
- $$z + \bar{z} \leq 6 \quad \text{and} \quad |z - 2| + |z - 4| \leq 10$$
- c) Find the perimeter of the shape in the Argand Diagram described by 2
- $$|z - 1| \leq 1 \quad \text{and} \quad 0 \leq \arg z \leq \frac{\pi}{6}$$

Question 9 (10 marks) Start a new page

- a) Find the equation of the tangent to $\frac{x^2}{16} + \frac{y^2}{25} = 1$ at the point $P(4 \cos \theta, 5 \sin \theta)$. 2

- b) $P(2p, \frac{2}{p})$ is a variable point on the hyperbola $xy=4$.

The normal to the hyperbola at P meets the hyperbola again at $Q(2q, \frac{2}{q})$.

M is the midpoint of PQ.

- (i) Show that the equation of the normal at P is given by $p^3x - py = 2(p^4 - 1)$ 2

- (ii) Show that $q = -\frac{1}{p^3}$ 1

- (iii) Show that M has coordinates $[\frac{1}{p}(p^2 - \frac{1}{p^2}), p(\frac{1}{p^2} - p^2)]$ 2

- (iv) Show that, as P moves on the curve $xy = 4$, the locus of M is given by 3

$$(x^2 - y^2)^2 = -x^3y^3$$

End of Examination

SOLUTIONS S.T.H.S. YR12 EXT 2. ASS. 2 MAR 2015

SECTION I

1. C 2. B 3. A 4. B 5. C

5x1 = 5 MARKS

SECTION II

6 a) 1) $x = 2\cos\theta$ + $y = \sqrt{2}\sin\theta$
 $x^2 = 4\cos^2\theta$ $y^2 = 2\sin^2\theta$

$\therefore \frac{x^2}{4} + \frac{y^2}{2} = \cos^2\theta + \sin^2\theta$
 $\quad \quad \quad = 1$ as req'd

$\therefore E$ can be written parametrically as (2)
 $x = 2\cos\theta$ $y = \sqrt{2}\sin\theta$

(ii) If $x = 2\cos\theta$ + if $y = \sqrt{2}\sin\theta$
 $\frac{dx}{d\theta} = -2\sin\theta$ $\frac{dy}{d\theta} = \sqrt{2}\cos\theta$

$\therefore \oint = 2 \int_0^\pi \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$ becomes

$= 2 \int_0^\pi \sqrt{4\sin^2\theta + 2\cos^2\theta} d\theta$

$= 2 \int_0^\pi \sqrt{4(1-\cos^2\theta) + 2\cos^2\theta} d\theta$

$= 2 \int_0^\pi \sqrt{4 - 2\cos^2\theta} d\theta$ (2)

$= 2\sqrt{2} \int_0^\pi \sqrt{2 - \cos^2\theta} d\theta$ as req'd

$$\begin{aligned}
 \text{b) i) } w &= \frac{1}{2}(1+i\sqrt{3}) \\
 \therefore w^3 &= \frac{1}{8}(1+i\sqrt{3})^3 \\
 &= \frac{1}{8}(1+i\sqrt{3})(1+i\sqrt{3})^2 \\
 &= \frac{1}{8}(1+i\sqrt{3})(-2+2i\sqrt{3}) \\
 &= -\frac{2}{8}(1+i\sqrt{3})(1-i\sqrt{3}) \\
 &= -\frac{1}{4} \times 4 \\
 &= \underline{\underline{-1}} \text{ as req'd}
 \end{aligned}$$

(1)

$$\begin{aligned}
 \text{ii) } w^{12} &= (w^3)^4 \\
 &= (-1)^4 \\
 &= \underline{\underline{1}}
 \end{aligned}$$

(1)

(iii) Cube roots of -1 are solutions of $w^3 = -1$

$$\text{Let } w = \cos \theta + i \sin \theta$$

$$\therefore w^3 = \cos 3\theta + i \sin 3\theta$$

$$\text{Thus } \cos 3\theta + i \sin 3\theta = -1 \text{ w } 0 \leq 3\theta \leq 6\pi$$

$$\therefore 3\theta = \pi, 3\pi, 5\pi$$

$$\theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$$

$$\text{Thus } w_1 = \underline{\underline{\cos \frac{\pi}{3}}} \text{ or } \underline{\underline{\frac{1}{2} + i\frac{\sqrt{3}}{2}}} \text{ (as given)}$$

$$w_2 = \underline{\underline{\cos \pi}} \text{ or } \underline{\underline{-1}} \text{ (real)}$$

$$w_3 = \underline{\underline{\cos \frac{5\pi}{3}}} \text{ or } \underline{\underline{\frac{1}{2} - i\frac{\sqrt{3}}{2}}}$$

(2)

c) If $3+2i$ is a root, then $3-2i$ is also a root

$\therefore (x-3-2i)(x-3+2i)$ is a factor

$\therefore (x-3)^2+4$ " " "

$\therefore x^2-6x+13$ " " "

Using long division

$$\begin{array}{r} x^2 + x - 2 \\ x^2 - 6x + 13 \overline{) x^4 - 5x^3 + 5x^2 + 25x - 26} \\ \underline{x^4 - 6x^3 + 13x^2} \\ x^3 - 8x^2 + 25x \\ \underline{x^3 - 6x^2 + 13x} \\ -2x^2 + 12x - 26 \\ \underline{-2x^2 + 12x - 26} \\ 0 \end{array}$$

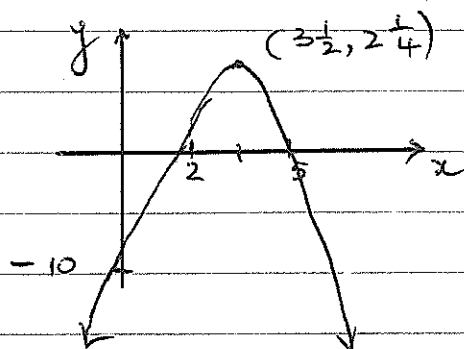
$$\begin{aligned} \therefore P(x) &= (x^2 - 6x + 13)(x^2 + x - 2) \\ &= (x - 3 - 2i)(x - 3 + 2i) : (x + 2)(x - 1) \end{aligned}$$

$\therefore P(x) = 0$ has solutions
 $3+2i, 3-2i, -1, 1$

2

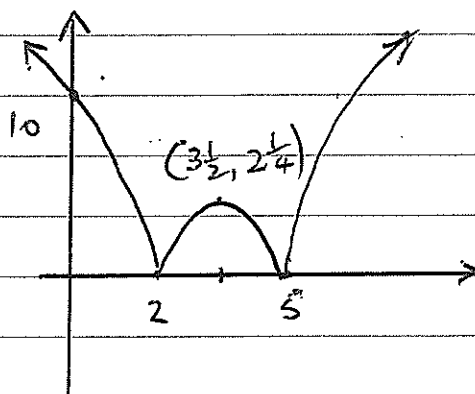
7. a) $f(x) = -x^2 + 7x + 10$
 $= -(x^2 - 7x + 10)$
 $= -(x-2)(x-5)$

(i) $y = f(x)$



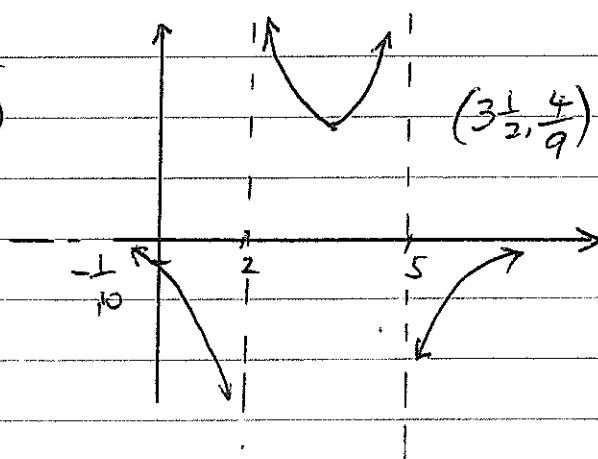
(1)

(ii) $y = |f(x)|$



(2)

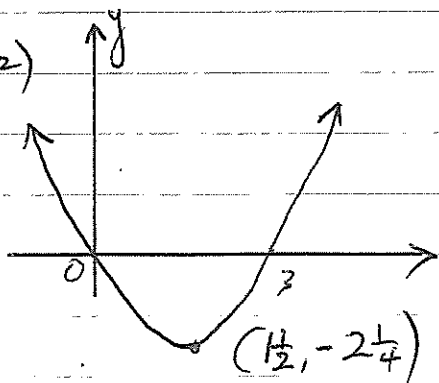
(iii) $y = \frac{1}{f(x)}$



(2)

(iv)

$y = -f(x+2)$



(2)

$$(b) \quad P(x) = 18x^3 + 3x^2 - 28x + 12$$

$$\text{Solve } P'(x) = 54x^2 + 6x - 28 = 0$$

$$\therefore 27x^2 + 3x - 14 = 0$$

$$\therefore x = \frac{-3 \pm \sqrt{9 + 4 \times 14}}{54}$$

$$= \frac{-3 \pm 39}{54}$$

$$= \frac{36}{54} \text{ or } \frac{-42}{54}$$

$$= \frac{2}{3} \text{ or } -\frac{7}{9}$$

So one of these is a repeated root of $P(x)$

$$P\left(\frac{2}{3}\right) = 0$$

$\therefore x = \frac{2}{3}$ is a double root of $P(x)$

$\therefore (3x-2)^2$ is a factor

$\therefore 9x^2 - 12x + 4$ is a factor

$$\begin{array}{r} 9x^2 - 12x + 4 \overline{) 18x^3 + 3x^2 - 28x + 12} \\ \underline{18x^3 - 24x^2 + 8x} \\ 27x^2 - 36x + 12 \\ \underline{27x^2 - 36x + 12} \\ 0 \end{array}$$

$$\therefore P(x) = (3x-2)^2(2x+3)$$

which has roots

$$\underline{\underline{\frac{2}{3} \text{ or } -\frac{3}{2} \text{ ONLY}}}$$

(3)

Q8

$$\begin{aligned} a) \quad z_1 &= 1 + i\sqrt{3} \\ &= 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \\ &= 2 \operatorname{cis} \frac{\pi}{3} \end{aligned}$$

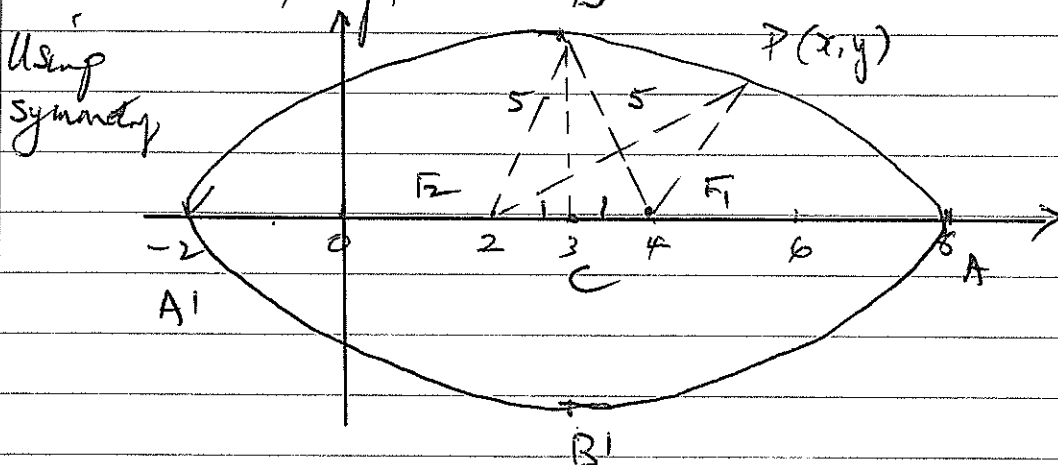
$$\begin{aligned} z_2 &= 1 - i \\ &= \sqrt{2}\left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right) \\ &= \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \end{aligned}$$

$$\therefore \frac{z_1}{z_2} = \frac{2 \operatorname{cis} \frac{\pi}{3}}{\sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)}$$

$$= \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right) \quad \text{IN MOD-ARG FORM}$$

(2)

b) i) Given $\left| \frac{z-2}{z-4} \right|$ is distance from z to $x=2$ } in Argand diagram
 " " " " " to $x=4$ }



Since P is on curve such that $F_1P + F_2P = 10$

$\therefore A$ must be $(8,0)$ A' must be $(-2,0)$

Length of major axis is $AA' = 10$ units

Centre is at C which must be $(5,0)$

$$BC^2 + 1^2 = 5^2$$

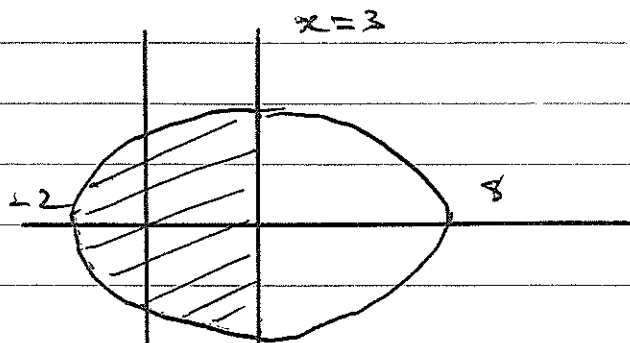
$$\begin{aligned} \therefore BC &= \sqrt{24} \\ &= 2\sqrt{6} \end{aligned}$$

$\therefore$ Length of minor axis is $BB' = 4\sqrt{6}$

(3)

(ii) $z + \bar{z} \leq 6$
 is $2x \leq 6$
 $x \leq 3$

& $|z-2| + |z-4| \leq 10$
 is region inside ellipse
 above:

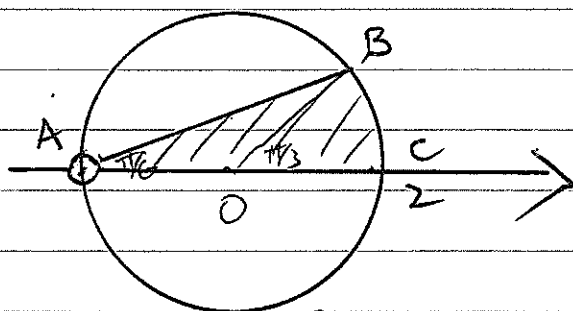


(3)

(c) $|z-1| \leq 1$ is region inside circle
 centre (1,0) radius 1

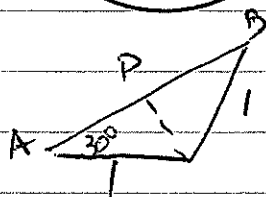
& $0 \leq \arg z \leq \pi/6$ as shown

[Actually $0 < \arg z \leq \pi/6$]



Shape ABC

$\triangle ABD$



P is midpoint of AD
 $AP = \cos 30^\circ = \frac{\sqrt{3}}{2}$

$\therefore AB = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$

Arc BC = $l = r\theta$
 $= 1 \times \pi/3$
 $= \frac{\pi}{3}$

$\therefore P = AB + \text{arc BC} + 2$
 $= \sqrt{3} + \frac{\pi}{3} + 2$ units

(2)

9. a) $\frac{x^2}{16} + \frac{y^2}{25} = 1$ Diff. implicitly

$$\frac{2x}{16} + \frac{2y}{25} \frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = -\frac{25x}{16y}$$

$$\therefore m_T = -\frac{25 \cdot 4 \cos \theta}{16 \cdot 5 \sin \theta}$$

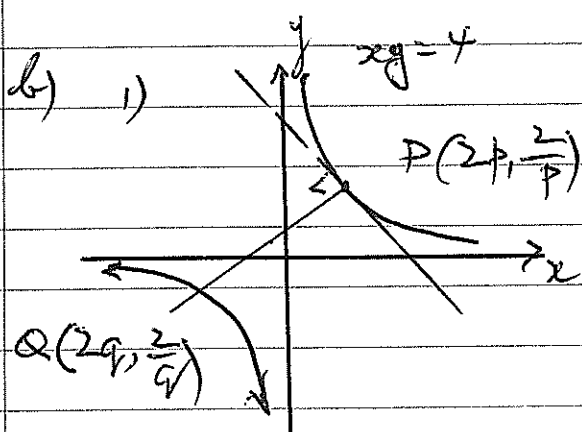
$$= -\frac{5 \cos \theta}{4 \sin \theta}$$

Eqn of tang is $y - 5 \sin \theta = -\frac{5 \cos \theta}{4 \sin \theta} (x - 4 \cos \theta)$

$$4 \sin \theta y - 20 \sin^2 \theta = -5 \cos \theta x + 20 \cos^2 \theta$$

$$\text{OR } \frac{\cos \theta}{4} x + \frac{\sin \theta}{5} y = 1$$

(2)



$$\begin{aligned} xy &= 4 \\ y &= \frac{4}{x} \\ \therefore \frac{dy}{dx} &= -\frac{4}{x^2} \end{aligned}$$

at P $m_T = -\frac{4}{(2p)^2} = -\frac{1}{p^2}$

$$\therefore m_N = +p^2$$

Eqn of Normal is

$$y - \frac{2}{p} = p^2 (x - 2p) \quad \text{--- (1)}$$

① $\times p$

$$py - 2 = p^3 x - 2p^4$$

OR $p^3 x - py = 2(p^4 - 1)$ as req'd

(2)

(ii) Reverting to ① & substituting
 $Q(2q, \frac{2}{q})$

$$\frac{2}{q} - \frac{2}{p} = p^2(2q - 2p)$$

$$\frac{1}{q} - \frac{1}{p} = p^2(q - p)$$

$$\frac{(p - q)}{pq} = -p^2(p - q)$$

* Noting $p \neq q$

$$\frac{1}{pq} = -p^2$$

$$\text{OR } q = -\frac{1}{p^3} \text{ as req'd}$$

①

(iii) Image of PQ is $M\left(\frac{2p+2q}{2}, \frac{\frac{2}{p} + \frac{2}{q}}{2}\right)$

$$\text{OR } \left(\frac{p+q}{1}, \frac{\frac{1}{p} + \frac{1}{q}}{1}\right)$$

$$\text{But } q = -\frac{1}{p^3} \therefore M\left(p - \frac{1}{p^3}, \frac{1}{p} - p^3\right)$$

②

$$M \text{ is } \left[\frac{1}{p}(p^2 - \frac{1}{p^2}), p\left(\frac{1}{p^2} - p^2\right)\right]$$

(iv) Checking $(x^2 - y^2)^2 = -x^3 y^3$

$$\begin{aligned} \text{L.H.S} &= \left[\frac{1}{p^2} \left(p^2 - \frac{1}{p^2} \right)^2 - p^2 \left(\frac{1}{p^2} - p^2 \right)^2 \right]^2 \\ &= \left[\frac{1}{p^2} \left(\frac{1}{p^2} - p^2 \right)^2 - p^2 \left(\frac{1}{p^2} - p^2 \right)^2 \right]^2 \\ &= \left[\left(\frac{1}{p^2} - p^2 \right)^2 \left(\frac{1}{p^2} - p^2 \right) \right]^2 \\ &= \left(\frac{1}{p^2} - p^2 \right)^6 \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= - \frac{1}{p^3} \left(p^2 - \frac{1}{p^2} \right)^3 \cdot \cancel{p^3} \left(\frac{1}{p^2} - p^2 \right)^2 \\ &= + \left(\frac{1}{p^2} - p^2 \right)^3 \left(\frac{1}{p^2} - p^2 \right)^3 \\ &= \left(\frac{1}{p^2} - p^2 \right)^6 \end{aligned}$$

As L.H.S = R.H.S we have

Confirmed Locus of M is

$$(x^2 - y^2)^2 = -x^3 y^3$$

(3)